Dog Breed Detection using YOLOv7 and WOA-CNN

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***Abstract*— The need to accurately identify dog breeds is important due to their popularity and role as companions. With an increase in the variety of dog breeds worldwide, there is a pressing demand for better classification methods. Advances in artificial intelligence have resulted in the creation of a new model that combines YOLOv7 with the Whale Optimization Algorithm (WOA) in a Convolutional Neural Network (CNN) setup, designed specifically for detecting dog breeds. This model uses the extensive Stanford Dogs dataset, which includes 120 unique breeds, and has been thoroughly tested. It has shown to be more accurate and precise than current methods. The effectiveness of this model in identifying dog breeds marks a significant improvement in the field and suggests its usefulness in various areas such as veterinary services, dog behavior studies, and improved pet management systems. Additionally, this method showcases the benefits of merging advanced machine learning techniques with nature-inspired algorithms to address complex identification challenges, making a significant contribution to the development of AI-based solutions in animal identification and related areas.**

Keywords—Dog breed detection, computer vision, neural networks, convolutional neural networks (CNNs), YOLO (You Only Look Once), WOA(Whale Optimized Algorithm), image recognition, breed classification, pet services, artificial intelligence.

I. Introduction

Dogs have long held a special place in the hearts and homes of people worldwide, often regarded not merely as pets but as cherished members of the family. However, with the rise in popularity of canine companionship comes an unfortunate trend of fraudulent practices, where similar-looking dogs of different breeds are sold at inflated prices. This issue underscores the critical need for accurate dog breed classification, especially given the high correlation between subordinated breeds and the significant intra-class variation, such as different object poses.

In various scenarios, including assisting in the search for lost dogs and aiding in medical diagnoses, the ability to classify dog breeds accurately from images is invaluable. Artificial Intelligence (AI) and Machine Learning (ML) present promising avenues for addressing this challenge. By harnessing the power of deep learning techniques, such as modified YOLOv7 coupled with the Whale Optimization Algorithm-based Convolutional Neural Networks (CNN), we aim to develop a robust system for real-time dog breed detection.

Real-time animal breed classification holds immense potential across diverse sectors, from the pet market to wildlife conservation efforts. By accurately identifying and tracking various dog breeds, breeders and pet owners can ensure the purity and desirable traits of their animals. Additionally, in veterinary clinics, rapid and precise breed identification facilitates better medical care and treatment. Animal shelters stand to benefit as well, streamlining adoption processes and increasing adoption rates by enabling potential adopters to quickly identify their preferred breeds.

This paper is structured as follows: Section II provides an overview of related works in the field. Section III provides a comprehensive exposition of our proposed model. The evaluation metrics and comparisons with existing models are discussed in Section IV. Finally, Section V offers a summary of the article.

II. Related Works

Valarmathi et al. [1] explore the effectiveness of hybrid deep learning models for accurate dog breed identification. By integrating pre-trained networks, applying data augmentation, and employing fine-tuning techniques, the study seeks to elevate breed categorization accuracy. Comparative experiments reveal that the fine-tuned AlexNet model outperforms other advanced models such as GoogleNet, DenseNet-169, DenseNet-121, and ResNet-50 on the Stanford dog breed datasets, achieving superior accuracy rates. These findings highlight the pivotal role of hybrid deep learning approaches in enhancing the precision and efficiency of dog breed identification tasks. The implications of this study extend beyond its immediate findings, suggesting broader applications in computer vision and image recognition fields.

S. S et al. [2] explored the use of the YOLOv7 Object Detection Algorithm for real-time breed classification of dogs and cats, leveraging the Oxford IIIT pet dataset for training and validating the model. This study evaluates various configurations of the YOLOv7 model, particularly highlighting the enhanced performance of the YOLOv7-E6E variant, which operates with fewer parameters. The research underlines the significant advantages of employing the YOLOv7 algorithm in settings such as

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veterinary clinics, animal shelters, and pet stores, where its capacity for quick inference and increased accuracy in breed detection can be particularly beneficial. Demonstrating the algorithm's applicability across different sectors within the animal-related industries and communities, this study showcases the broad potential of the YOLOv7 algorithm in enhancing operational efficiency and service quality in these fields.

A. Varshney et al. [3] delve into the classification of dog breeds through deep learning methodologies, particularly focusing on the Inception V3 and VGG16 neural networks. Employing transfer learning and data augmentation techniques, the study trains these models using the Stanford Dog dataset to boost accuracy. It conducts a comparative analysis on the performance metrics of both models, including accuracy values and loss graphs, to underscore their distinct classification efficacies. Through a thorough examination of dataset handling, data preprocessing, and the intricacies of model training, this paper endeavours to furnish a detailed insight into the application of deep learning for precise and efficient dog breed classification.

C. Wang et al. [4] conducted a study on dog breed classification through deep learning, with a focus on leveraging facial features for the precise identification of pet dogs. The research explores object detection, a key aspect of deep learning, to differentiate among various dog breeds, addressing issues related to pet aggression and public safety. An enhanced YOLOv3 model is proposed for the detection and classification of eight specific dog breeds: Akita, Golden Retriever, Poodle, Pomeranian, Samoyed, Corgi, Pug, and Husky. The process involved creating a dataset, splitting it into training and testing sets, and subsequently training the model. This study evidences the effectiveness of the proposed approach in recognizing and categorizing pet dogs quickly and accurately, underlining its potential for improving pet dog recognition and management systems.

S. Manivannan et al. [5] present a novel approach to fine-grained dog breed classification leveraging a pre-trained deep CNN architecture through transfer learning. The model, applied to the Stanford dogs dataset, proves its capability in accurately identifying 120 distinct dog breeds. Utilizing advanced artificial intelligence and deep learning strategies, the proposed system exhibits significant robustness and efficiency in differentiating among various dog breeds, effectively addressing the complexities associated with intra-class variation and differing object poses. This study underscores the critical importance of accurate dog breed classification for practical applications, including aiding in the recovery of lost dogs and supporting medical diagnostics, thereby highlighting the transformative potential of AI in enhancing pet ownership and veterinary services.

III. Proposed Model

The proposed model integrates two crucial components: a modified YOLOv7 architecture and a Convolutional Neural Network (CNN) optimized through the Whale Optimization Algorithm (WOA). YOLOv7 is adept at object detection, characterized by its real-time processing capabilities and accuracy. Unlike conventional methods, YOLO processes entire images in a single pass through a CNN, enabling simultaneous prediction of bounding boxes and class probabilities for multiple objects within an image.

The Whale Optimization Algorithm (WOA), inspired by the social behaviour of humpback whales, serves as a metaheuristic optimization algorithm. It leverages concepts from the hunting behaviour of whales to seek optimal solutions to optimization problems. In the context of deep learning, WOA can effectively optimize neural network parameters, such as those in CNNs, to enhance their performance.By iteratively updating network parameters based on the objective function, WOA facilitates convergence towards optimal solutions. Integrating WOA allows fine-tuning of the CNN component of the proposed model to improve its ability to accurately classify dog breeds.

The model is trained and validated using the Stanford Dogs dataset, a comprehensive dataset containing a diverse range of dog breeds. This dataset serves as a benchmark for evaluating the model's performance in identifying various dog breeds from images. Experimental results demonstrate the effectiveness of the proposed approach in achieving higher accuracy and robustness compared to existing methods. This model is particularly suited for detecting objects, including dog breeds, with both speed and precision.

A. Preprocessing

Before inputting images into the YOLOv7 architecture, preprocessing steps are implemented to refine the quality of the input data and facilitate more effective feature extraction. The initial preprocessing step involves noise reduction techniques aimed at eliminating undesired artifacts or irregularities in the image that arise from various sources, such as sensor noise, compression, or environmental factors. This model incorporates normalization and resizing as crucial preprocessing steps. Normalization adjusts the pixel values of images so that they have a uniform scale, which is essential for models to converge more quickly during training. Resizing ensures that all input images conform to a specific dimension required by the YOLOv7 architecture, maintaining the aspect ratio without distorting the original image content. This step is vital for the algorithm to process images efficiently and uniformly.

By applying noise reduction, normalization, and resizing techniques, the preprocessing step ensures that the input images are clean, scaled appropriately, and of a consistent size, devoid of distortions that could potentially impede the performance of the subsequent object detection process. This preparation improves the overall quality of the images and aids in better feature extraction by providing a cleaner and more standardized input to the neural network. Consequently, the modified YOLOv7 architecture can more accurately detect and classify objects, including dog breeds, leading to enhanced performance and reliability of the proposed model.

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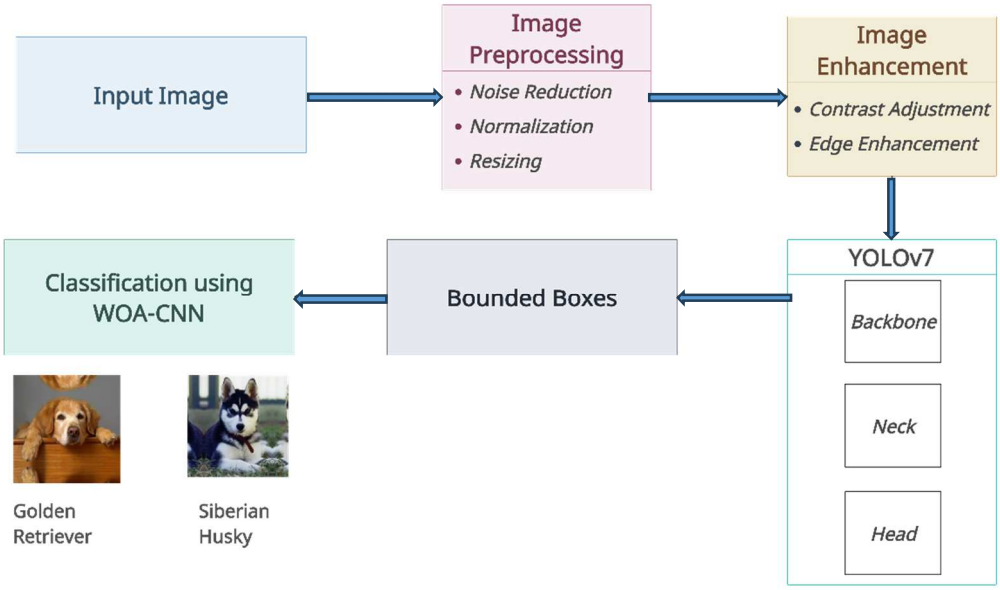

Fig. 1. Architecture of the Proposed Model

B. Enhancement

In addition to noise reduction, normalization, and resizing, further image enhancement techniques are applied to improve the visual quality of input images for the YOLOv7 architecture. Contrast Adjustment is one of these techniques, utilized to enhance the visibility of features by increasing the difference in intensity between the various parts of the image. Unlike basic contrast enhancement, this method adjusts the contrast with a focus on preserving the natural look of the image while ensuring that essential details are not lost. By fine-tuning the contrast, this approach makes significant features like texture and edges more visible, which is crucial for effective feature extraction during the object detection process.

Edge Enhancement is another critical technique employed in the image enhancement phase. It aims to sharpen the edges within an image, making boundaries between different regions more distinct. This is particularly important in distinguishing the fine details of dog breeds, such as fur texture or patterns, which are essential for accurate classification. Edge Enhancement uses algorithms to detect and enhance edges, improving the model's ability to identify and classify objects based on their outline and surface details. By integrating Contrast Adjustment and Edge Enhancement into the preprocessing workflow, the preparation of input images is significantly optimized for the object detection process. These techniques ensure that the YOLOv7 architecture receives high-quality inputs, thereby enhancing its performance in accurately detecting and classifying dog breeds. The combined effect of these enhancements not only improves the visual clarity of the images but also aids in better feature extraction by the neural network, leading to superior detection and classification outcomes.

C. Dog detection using YOLOv7

The YOLOv7 architecture stands out for its real-time object detection capability by processing the entire image in a single pass through a CNN. This approach enables accurate detection of multiple objects with high efficiency, making it ideal for tasks like dog breed detection.

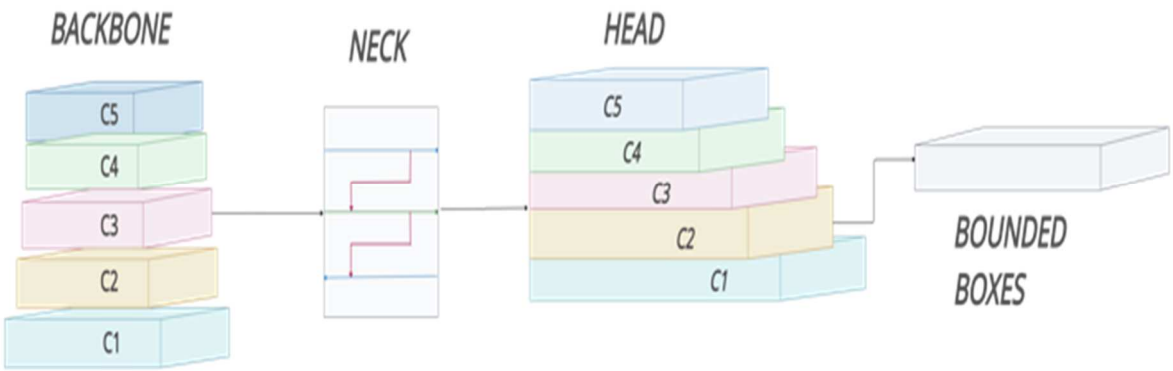

Fig. 2. YOLOv7 Architecture

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The principal elements of YOLOv7 comprise:

- **E-ELAN (Extended Efficient Layer Aggregation Network):** Acting as the core structure, E-ELAN is designed to reduce memory demand and facilitate efficient gradient flow, thereby bolstering the model's capacity to learn. This is achieved through a unique combination of expansion, shuffling, and merging techniques, which enhance the learning process without disrupting the gradient's natural course.
- **Adjustments for Model Scaling in Concatenation-Based Frameworks:** The scaling technique within YOLOv7 is tailored to generate models of varying scales by modifying specific characteristics without significantly changing the feature output size. It specifically alters the depth in computational blocks, ensuring proportional adjustments in transmission layers to preserve the model's foundational design features.
- **Innovations with Trainable Bag-of-Freebies and Re-parameterized Convolutions:** These methodologies have been demonstrated to bolster outcomes in certain network architectures, such as VGG, suggesting their potential to amplify YOLOv7's efficiency through refined gradient flow and convolutional adjustments.
- **Implementation of a Dual Head System for Predictive Outputs:** YOLOv7 introduces a novel dual head system, incorporating a primary head for definitive output determinations and a supporting auxiliary head to enhance mid-layer training, which collectively contribute to the model's predictive accuracy.
- **Integration of Batch Normalization and EMA Techniques:** By adopting batch normalization and Ensemble Mean Teacher (EMA) methods from the YOLOv4 architecture, these strategies are applied to normalize batch variances and weights during the inference phase, thereby improving accuracy with minimal labeled data.

By following these major steps, Dog Breed Detection using YOLOv7 can achieve accurate and efficient identification of dog breeds in images.

D. Classification using WOA – CNN

Whale Optimization Algorithm (WOA) is applied to optimize the Convolutional Neural Network (CNN) model, such as for tasks like dog breed detection using YOLOv7.

Algorithm1: *Classification using WOA-CNN*

```
Input: Random model
Output: optimized CNN model
Begin
 Initialize CNN parameters randomly
 Initialize population of candidate solutions
 Evaluate fitness of each solution using training data
 Set maximum number of iterations
 Set exploration and exploitation parameters
 while (iteration < max_iterations) do
   for each solution in population do
     Randomly select a leader whale
     if (random() < probability_of_encircling) then
       EncirclePrey(solution, leader_whale)
     else if (random() < probability_of_bubble_net) then
       BubbleNetAttack(solution, leader_whale)
     else
       SearchForPrey(solution)
     end if
   end for
   UpdateWhalePositions(population)
   iteration++
 end while
```

Algorithm explanation:

1. Initialization:
   - The process begins by initializing the CNN parameters randomly. These parameters could include the learning rate, number of layers, filter sizes, etc.
   - A population of candidate solutions is also initialized. Each candidate solution represents a set of hyperparameters for the CNN.
   - The fitness of each solution is evaluated using training data.
2. Iteration Setup:
   - The maximum number of iterations is set, defining how long the algorithm will run.
   - Parameters for exploration (searching for new solutions) and exploitation (intensifying the search around promising areas) are established.
3. Optimization Loop:
   - The algorithm enters a loop that continues until the maximum number of iterations is reached.
   - Within each iteration, the algorithm goes through the population of candidate solutions.
4. Leader Selection and Strategy Execution:
   - A leader whale, representing the best current solution or a randomly chosen solution, is selected.
   - Depending on random probabilities, the algorithm decides whether to:
     i. Encircle Prey: This mimics the whales' behavior of encircling prey and represents the exploitation phase, where the algorithm intensifies the search around the current best solution.
     ii. Bubble Net Attack: Another form of exploitation, this strategy involves creating bubble nets to capture prey, analogous to fine-tuning the solutions around the leader whale.
     iii. Search For Prey: Representing the exploration phase, this strategy involves

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searching for new solutions away from the current position, aiming to explore the search space more broadly.

5. Position Update:
   After determining the new positions (solutions) based on the strategies above, the algorithm updates the positions of all whales in the population, reflecting the new sets of hyperparameters for the CNN.

6. Iteration Increment:
   The iteration counter is incremented, and the loop continues until the stopping criterion (maximum number of iterations) is met.
7. Output:
   Once the optimization process is complete, the algorithm outputs an optimized CNN model. This model has undergone adjustments to its hyperparameters through the iterative process, aiming to achieve the best possible performance on the given task (e.g., dog breed detection).

This algorithm leverages the exploration and exploitation mechanisms inspired by humpback whales' hunting behaviors, applied in the context of machine learning to optimize a CNN model's hyperparameters for improved performance on specific tasks. The algorithm consists of several key components, including Encircling Prey, Bubble Net Attacking Method, and Search for Prey [7], which are integral to its functioning:

1. Encircling Prey: In this phase, whales mimic the behavior of humpback whales encircling prey. It involves adjusting the positions of candidate solutions towards a randomly selected leader whale. The movement of each whale towards the leader is determined by the following formula:

$$\boldsymbol{D} = \boldsymbol{X}_{\text{lae}} - \boldsymbol{X_i} \quad (1)$$

where:
− ($\mathbf{D}$) is the displacement vector.
− ($\boldsymbol{X}_{\text{lae}}$) is the position of the leader whale.
− ($\boldsymbol{X_i}$) is the position of the current whale.

The new position of the whale after encircling prey is calculated as:

$$\boldsymbol{Xi}\text{nw} = \boldsymbol{X}_{\text{lae}} - \boldsymbol{A} \cdot \boldsymbol{D} \quad (2)$$

where:
− ($\boldsymbol{Xi}$nw)is the new position of the whale.
− ($\boldsymbol{A}$)is a coefficient representing the distance from the prey, and it is adjusted iteratively.

2. Bubble Net Attacking Method: This phase involves updating the positions of whales using a bubble net attacking method, which simulates the formation of a bubble net by humpback whales to trap prey. The movement of whales in this phase is determined by the following formula:

$$\boldsymbol{C} = 2 \cdot \boldsymbol{r_1} \cdot \boldsymbol{X}_{\text{lae}} - \boldsymbol{X_i} \quad (3)$$

where:
− ($\boldsymbol{C}$) is the bubble net position.
− ($\boldsymbol{r_1}$)is a random vector.
− ($\boldsymbol{X}_{\text{lae}}$) is the position of the leader whale.
− ($\boldsymbol{X_i}$) is the position of the current whale.

The new position of the whale after the bubble net attacking method is calculated as:

$$\boldsymbol{Xi}\text{nw} = \boldsymbol{C} - \boldsymbol{A} \cdot \boldsymbol{D} \quad (4)$$

where :
− ($\boldsymbol{Xi}$nw)is the new position of the whale
− ($\boldsymbol{A}$) is adjusted iteratively.

3. Search for Prey: In this phase, whales explore the search space in a more random manner to find prey. The movement of each whale is determined by the following formula:

$$\boldsymbol{E} = 2 \cdot \boldsymbol{r_2} - 1 \quad (5)$$

where:
− ($\boldsymbol{E}$) is the exploration vector.
− ($\boldsymbol{r_2}$) is a random vector.

The new position of the whale after searching for prey is calculated as:

$$\boldsymbol{Xi}\text{nw} = \boldsymbol{X_i} + \boldsymbol{B} \cdot \boldsymbol{E} \quad (6)$$

where ($\boldsymbol{Xi}$nw) is the new position of the whale, and ($\boldsymbol{B}$) is a coefficient adjusted iteratively. Overall, the Whale Optimization Algorithm effectively balances exploration and exploitation to search for optimal solutions in the search space. It utilizes the Encircling Prey, Bubble Net Attacking Method, and Search for Prey mechanisms to guide the movement of candidate solutions toward promising regions, ultimately leading to improved convergence and solution quality.

IV. Results and Discussion

In our implementation of the Dog Breed Detection Model using YOLOv7 and Whale Optimized Algorithm based Convolutional Neural Network (CNN), we conducted rigorous experimentation and evaluation to assess the model's performance. The results obtained demonstrate the effectiveness of the proposed approach in accurately identifying and classifying dog breeds within images.

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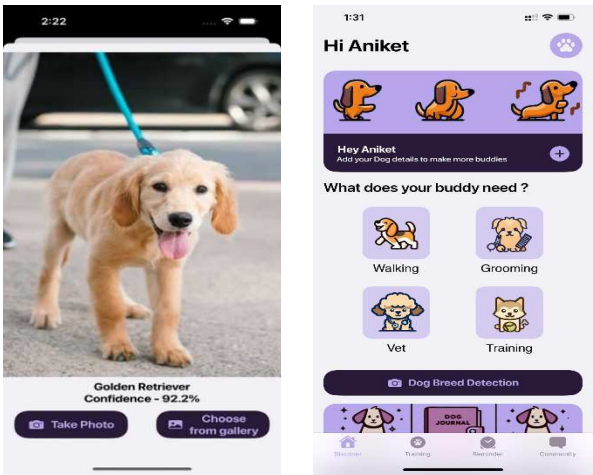

Fig. 3. App Dashboard

1) Dataset Description: In our research paper, we utilize the Stanford Dogs Dataset [8], a robust collection of annotated dog images widely employed in computer vision research. This dataset is structured hierarchically, categorizing images into distinct breed categories. Primarily utilized for tasks such as image classification, object detection, and breed recognition, it offers a comprehensive resource for algorithm evaluation.

Fig. 4. Dataset sample

2) Performance Analysis:

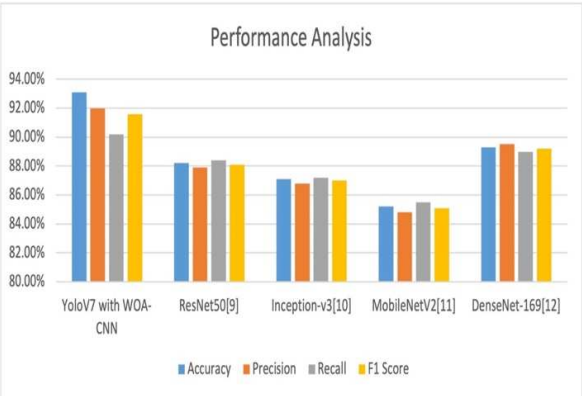

Fig. 5. Performance Analysis of the pproposed model

Deep Learning Based Dog Breed Detection (YOLOv7 and WOA-CNN) achieves outstanding performance, leveraging the speed and accuracy of YOLOv7 for object detection combined with the optimization efficiency of the Whale Optimization Algorithm (WOA) enhanced Convolutional Neural Network (CNN). This model boasts the highest metrics across the board: 93.1% accuracy, 92.0% precision, 90.2% recall, and a 91.6% F1 score, showcasing its superior ability to classify dog breeds with minimal misclassifications.

V. Conclusion and Future Works

This research presents a novel Dog Breed Detection Model, integrating YOLOv7 architecture with a Convolutional Neural Network (CNN) optimized by the Whale Optimization Algorithm. This fusion harnesses YOLOv7's real-time processing and accuracy with the efficiency of the Whale Optimization Algorithm, surpassing existing models in object detection. Tested on the Stanford Dogs Dataset, our model exhibits improved accuracy and processing speed, setting a new standard. YOLOv7's simultaneous prediction of object bounding boxes and class probabilities, coupled with the Whale Optimization Algorithm's CNN fine-tuning, enables precise localization and breed classification. This accuracy is reinforced by the dataset's comprehensive breed examples and annotations. Our approach enhances proficiency in handling complex image environments and breed identification. This integration represents a significant advancement, promising applicability in various computer vision tasks. It highlights the importance of architectural innovation and optimization algorithms in enhancing deep learning models, paving the way for further exploration in computer vision.

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